NASA Wildland Fire: Fires in the Southeastern US from 2017-2021

Avi Grant

Thank you to Rob Sohlberg, Dr. Joanne Hall and Dr. David Green

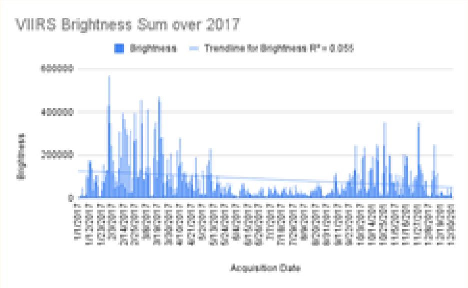
What I worked on this semester?

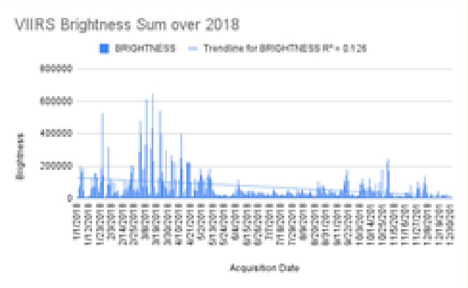
- I assisted in identifying fire-related use cases integrating multiple remote-sensing data sources for a literature review.
- I used ArcGIS to map fires in the Southeastern US in a temporal analysis from 2017-2021 to document changes in fire density and other statistics preceding and during the Coronavirus pandemic.

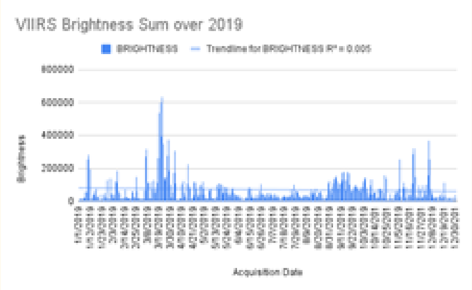
STANDARD VIIRS 375 m Active Fire product VNP14IMGTML distributed from NASA FIRMS. Available on-line https://firms.modaps.eosdis.nasa.gov/download/

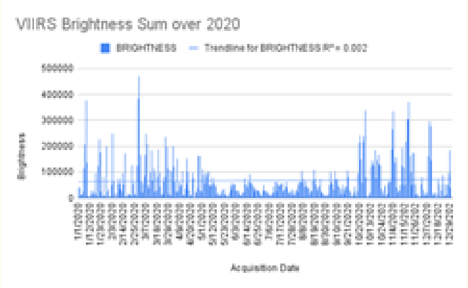
Even with year to year fire intervariability, we saw consistent clusters of burning in areas such as near New Orleans and the Florida panhandle near Tallahassee.

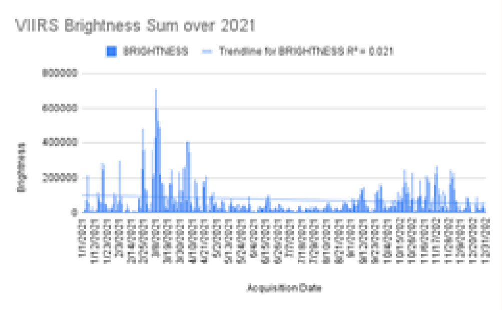
In addition, we located an anomaly along the research triangle region (consisting of Duke, the University of North Carolina at Chapel Hill, and North Carolina State University) where consistent fires would not usually be suspected. Thus we assume that the clusters of fire is caused by anthropogenic factors.











These are charts I made showing overall fire intensity by season using bimodal distribution in summed brightness year to year.

Lastly, we found that fires occur mostly in early spring and late fall of North American growing season, conveying that the driver of these fires is agriculture activity.